

Element H: Prototype Testing and Data Collection Plan

Testing Plan

Requirement	Testing Setup and Procedure	Test Cases and Number of Trials	Expected Range of Results	Pass/Fail Criteria, Goal
Functionality	Place the bookend on a flat surface. Apply an increasing number of books until the desired weight has been achieved. Observe if the bookend remains upright, stable, and can prevent the books from falling or sliding.	Test with book stacks of 1.680kg, 2.97kg, and 5.449kg, (put different weights) 3 trials per case for 9 trials in total	The bookend can hold books weighing 1.680kg To The bookend can hold books weighing 5.449kg	Pass: Bookend remains upright and can hold 1.680kg of weight Fail: Bookend is unstable, tips, slides, or can not hold 1.680kg of weight Goal: Bookend remains upright and can hold 5.449kg of weight
Sustainability	Evaluate the materials used in the bookend, check if they are recyclable, biodegradable, or reusable. Note the cost of the materials.	Break down the materials and classify the bookend as sustainable or not.	Must be within a budget of \$0.00. So completely made of reused materials.	Pass: 100% reused/materials we already have Fail: <100% reused/preowned Goal: 100% reused/preowned
Aesthetics	Have students evaluate the look of the bookend, giving a rating of 1 to 5.	Have 7 students give a rating	Colorful, creative, and original	Pass: 3 Fail: < 3 Goal: 5

Approval of Stakeholders (State the feedback 2 stakeholders gave and then how you incorporated it by showing the new testing table below) (Miracle)

- Mr. Eisenburg said to make a rubric for the students to go off of when rating the aesthetics
- Mr. Eisenburg said to talk everything over with the team
- A student who uses the library, Aneira, said we should have a rubric for the aesthetics and that we should make the expected range of results for aesthetics more detailed

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Aesthetics	Have students evaluate the look of the bookend,	Have 7 students give a rating Rating Rubric	Rating: 1 The bookend could be more	Pass: 3 Fail: < 3

	giving a rating of 1 to 5.	1: Plain or messy. Lacks color and design, looks incomplete or rushed 2: Decorated but still boring with minimal design, not eye-catching 3: Looks okay, has color and design, could be better 4: Neat and eye-catching. Creative with color and design, looks like it was well thought out 5: Outstanding. Extremely colorful and eye-catching. Professional and visually appealing. Very creative	colorful and creative. It lacks design and looks like thought wasn't put into it. To Rating 5 The bookend is very colorful, creative, and original. It catches someone's eye immediately and was created carefully and thoughtfully	Goal: 5
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